

YASSIN MOHAMED AWAD

Computer & Communication Engineering Student | Alexandria, Egypt

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PROFESSIONAL SUMMARY

Computer and Communication Engineering student (GPA 3.72/4.0) specializing in full-stack web development, computer vision, and robotics software engineering. Proficient in JavaScript, React, Node.js, PostgreSQL, Python, C, OpenCV, and ROS. Hands-on experience building scalable web applications, autonomous robotics systems, and real-time computer vision pipelines. Seeking a software engineering or full-stack development internship to apply technical expertise in production environments.

EDUCATION

Bachelor of Computer and Communication Engineering

Alexandria University

Expected Graduation: 2028

GPA: 3.72 / 4.0

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, SQL

Front-End Development: React, HTML5, CSS3, JavaScript (ES6+)

Back-End Development: Node.js, Express.js, RESTful APIs

Databases: PostgreSQL, SQL (relational schema design, CRUD operations)

Robotics & Automation: ROS (Robot Operating System), ROS2, autonomous navigation, robot control logic

Computer Vision: OpenCV, image processing, object detection, lane detection, facial recognition

Desktop Development: PyQt5 (GUI applications)

Tools & Concepts: Git, GitHub, OOP, Data Structures, Agile, Linux/Ubuntu

EXPERIENCE

Robotics Software Developer — Robotics Team

Alexandria, Egypt

- Developed computer vision modules for autonomous robots using Python and OpenCV, enabling real-time object detection and environmental perception.
- Integrated ROS (Robot Operating System) for inter-module communication, sensor data processing, and autonomous navigation workflows.
- Implemented robot control logic and motion planning algorithms for autonomous path following and obstacle avoidance.
- Built PyQt5 GUI desktop applications to visualize robot movement, sensor data streams, and playground maps in real time.

PROJECTS

Autonomous Robot System | *Python, ROS/ROS2, OpenCV, PyQt5*

- Designed and developed a fully autonomous robot system integrating ROS for modular communication between perception, planning, and control nodes.
- Implemented ROS topics, services, and launch files to coordinate sensor input, computer vision processing, and motor control in real time.
- Developed lane detection and obstacle avoidance modules using OpenCV, enabling the robot to navigate defined environments autonomously.
- Built a PyQt5 dashboard to monitor robot state, visualize sensor data, and display live map of robot trajectory.
- Performed system integration testing in simulated and physical environments to validate navigation accuracy and reliability.

Lane Detection System | *Python, OpenCV, ROS*

- Developed a real-time lane detection pipeline using OpenCV image processing techniques including Canny edge detection, Hough transform, and region-of-interest masking.
- Integrated the lane detection module as a ROS node, publishing lane position and deviation data to the robot control system via ROS topics.

- Optimized the computer vision pipeline for low-latency processing suitable for real-time autonomous vehicle navigation.
- Validated detection accuracy across varying lighting conditions and road surface types.

Face Recognition System | *Python, OpenCV, ROS*

- Built a face recognition system using OpenCV and deep learning-based feature extraction for real-time identity verification.
- Integrated the face recognition module as a ROS node, enabling the robot to recognize and respond to authorized personnel.
- Implemented face detection, alignment, and encoding pipeline achieving reliable recognition performance in varied lighting conditions.
- Published recognition events via ROS topics for downstream task execution and access control logic.

Perfume Store — Full-Stack E-Commerce Application | *JavaScript, React, Node.js, Express.js, PostgreSQL*

- Developed a full-stack e-commerce web application supporting product browsing, cart management, and order processing.
- Built RESTful APIs using Node.js and Express.js following REST best practices with proper HTTP status codes and error handling.
- Designed and managed a normalized relational database schema using PostgreSQL with optimized queries.
- Implemented a complete admin panel for managing products, orders, and users with role-based access control.
- Implemented JWT-based authentication and authorization for secure user sessions.

To-Do List — Full-Stack Task Management Application | *JavaScript, React, Node.js, Express.js, PostgreSQL*

- Developed a full-stack task management application with real-time CRUD operations via RESTful API.
- Designed PostgreSQL database schema with user-task relationships and ownership enforcement.
- Implemented user authentication and task ownership validation to ensure data isolation per user.

Product Management System — Front-End | *JavaScript, HTML5, CSS3*

- Developed a responsive front-end product management system with full CRUD functionality using vanilla JavaScript.
- Designed an interactive, mobile-friendly UI with dynamic DOM manipulation and event-driven architecture.

Bank Management System | *C Programming Language*

- Developed a console-based bank management system implementing account creation, deposit, withdrawal, and balance tracking.
- Applied structured programming, file I/O handling, and memory management principles in C.

CERTIFICATIONS

The Complete Full-Stack Web Development Bootcamp — Udemy

2025

[View Credential](#)

ACTIVITIES & INTERESTS

- Active member of university Robotics Team — contributed to autonomous robot design, computer vision, and ROS integration.
- Interests: Artificial Intelligence, Computer Vision, Autonomous Systems, Software Engineering, Full-Stack Web Development.